Geometry
Unit 6
Lesson 6 Practice

Name__Answer Key_______
Date___________________
Period__________________

1.
- o Only dilations
- o Only translations, reflections, and rotations
- ✖ All transformations

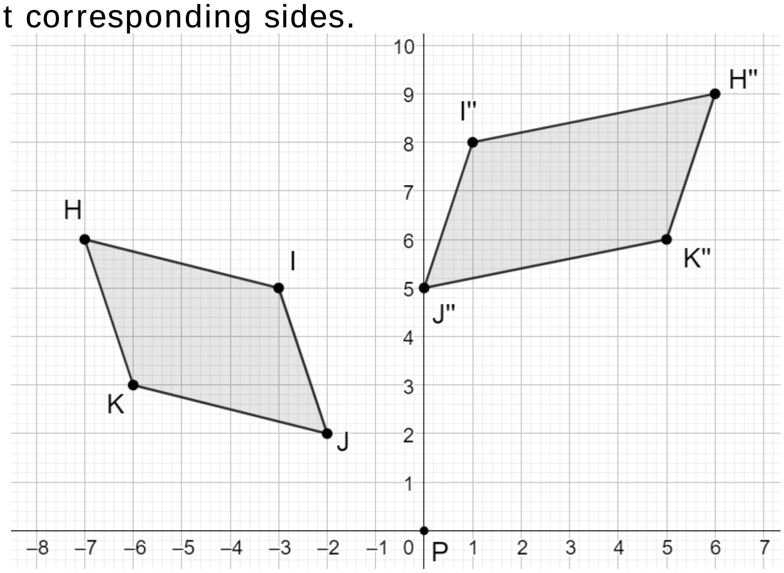
result in an image with congruent corresponding angles.

- o Only dilations
- ✖ Only translations, reflections, and rotations
- o All transformations

result in an image with congruent corresponding angles and sides.

- ✖ Only dilations
- o Only translations, reflections, and rotations
- o All transformations

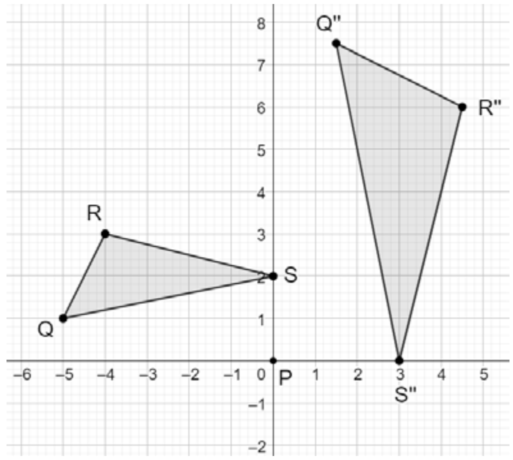
result in an image with proportional, but not congruent corresponding sides.



2. Write a sequence of transformations for preimage $HIJK$ that would show whether the figures are similar.
Figures are not similar.

3. Is $H''I''J''K''$ similar to the preimage?
- o Yes
- ✖ No

4. Explain the reason for your answer for #3.
Answers may vary. Not all of the points were transformed correctly. K'' should be at $(4,6)$ and H'' should be at $(5,9)$ in order to show the correct transformations preserving congruence and similarity from quadrilateral to quadrilateral.

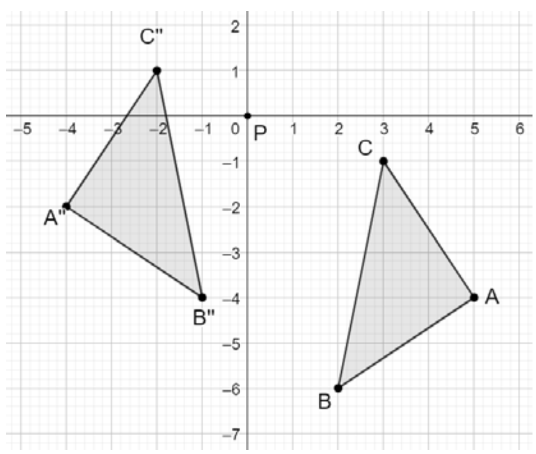


5. Write a sequence of transformations for preimage $\triangle QRS$ that would show whether the figures are similar.
Rotation $90°$ clockwise about the origin, then dilation from the origin with a scale factor of $\frac{3}{2}$.

6. Is $\triangle Q''R''S''$ similar to the preimage?

✖ Yes
o No

7. Explain the reason for your answer for #6.
Because $\triangle Q''R''S''$ was obtained by a sequence of transformations including a dilation.



8. Write a sequence of transformations for preimage $\triangle ABC$ that would show whether the figures are similar.
Reflection across the y-axis, then translation 1 unit right and 2 units up.

9. Is $\triangle A''B''C''$ similar to the preimage?

✖ Yes
o No

10. Explain the reason for your answer for #9.
Congruent figures are also similar, as you can use a dilation with a scale factor of 1.